Bahria University Lahore Campus

Department of Computer Sciences

Course Code/number	**GSC - 320**
Course Title/Name	**Numerical Analysis**
Credit Hours/Contact hours	3
Degree Program	Bachelor of Science in Computer Science (BSCS)
Prerequisites or Co-requisites	GSC-210 (Differential Equations)
Assessment Methods and Weightage	Quizzes 10 Assignments/Projects 20 Mid-Term Examination 20 Final Examination 50 **Total 100**
Textbook	Numerical Analysis by Burden, Richard L.cd J. Douglas Faires
Reference Material (With Edition, ISBN#)	1. Numerical Methods in Scientific Computing by Germund, D. Åke, B. 2. Numerical Methods for Scientific Computing by J. H. Heinbockel. 3. Numerical Analysis by I. A. Khubaza. 4. Numerical Analysis and Programming by Shan S Kuo.
Web Resources/ URL (if any)	
Instructor Name/Cluster Head/Subject Expert	Instructor Name: Zoobia Fatima Designation: Lecturer Status **Regular**/Visiting Cluster Head Name: Dr. Iram Noreen
Course Aims:	This course is designed as an introductory course in computational techniques for solving problems from science and engineering with emphasis on applications. The course will cover the classical fundamental topics in numerical methods such as, approximation, numerical integration, and numerical linear algebra, solution of nonlinear algebraic systems and solution of ordinary and partial differential equations.

Course Objectives	The primary objective of this course is to develop the basic understanding of the construction of numerical algorithms. This course will help students in building of numerical algorithms to estimate the solutions to common problems in science. The emphasis of the course will be the thorough study of numerical algorithms to understand the accuracy that various methods provide, the efficiency and precision for large scale systems, and issues of stability of methods.

Course Learning Outcomes	**After the successful completion of this course the students will learn**	**PLO**	**BT Level**
	1. **Demonstrate** the understanding of common numerical methods.	1	C2
	2. **Apply** typical numerical approaches, errors, to compute approximate solutions to Mathematical problems.	2	C3
	3. **Evaluate** numerical methods for a variety of Mathematical processes and problems, including integration and differentiation.	3	C5
	4. **Develop** numerical methods using a programming language.	2	C6

Course Description	It is well-known that the use of numerical methods for the analysis, simulation, and design of engineering processes and industrial systems has been increasing at a rapid rate. Therefore, this course is intended to prepare future engineers and computational scientists, in understanding the fundamentals of numerical methods, especially their application, limitations, and potentials.

Lecture Plan (16 Weeks)

Week # and Date	Lecture/ Hour #	Topic to be covered	Learning outcomes	Reference Text
1	1	Introduction to the course, mathematical preliminaries, error analysis, round off error	Understanding of the course objectives, and policies of the course regarding assessments, calculation of significant digits	Textbook and Lecture Handouts
	2	Solution of non-linear equation	Understanding errors in computing roots for system of non linear equations.	Textbook and Lecture Handouts
2	1	Solution by Bisection Method,	locating roots of nonlinear equations by bisection method	Textbook and Lecture Handouts
	2	Solution of non-linear equation by Regula Falsi Method	Locating roots of nonlinear equations by Regula-Falsi Method	Textbook and Lecture Handouts

	3	1	Solution of Non Linear Equations (Newton Raphson Method)	Locating roots of nonlinear equations by Newton Raphson method	Textbook and Lecture Handouts
		2	Solution of Non Linear Equations (Secant Method)	Locating roots of nonlinear equations by secant method.	Textbook and Lecture Handouts
	4	1	Solution of Non Linear Equations (Method of Iteration)	Locating roots of nonlinear equations by method of iterations	Textbook and Lecture Handouts
		2	Solution of Linear System of Equations(Jacobi Method),	Finding the solutions of systems of simultaneous linear equations	Textbook and Lecture Handouts
	5	1	Solution of Linear System of Equations(Gauss–Seidel Iteration Method)	Finding the solutions of systems of simultaneous linear equations	Textbook and Lecture Handouts
		2	Difference, Forward differences, Backward differences, difference table	Introduction to difference, and forward difference. An understanding of backward differences and difference table.	Textbook and Lecture Handouts
	6	1	Newton's differences formulas	A way of approximating a function with an nth degree polynomial passing through equally spaced points.	Textbook and Lecture Handouts
		2	Interpolation with equally spaced data , Newton forward Interpolation formula	A well-known approach to interpolation with equally spaced data.	Handouts, text book
	7	1	Newton backward interpolation formula	Finding the solution of any value of function from ending of the data.	Textbook and Lecture Handouts
		2	Gauss's forward interpolation formula, Gauss's Backward interpolation formula	The Gauss form of interpolation polynomial shows the linear character of polynomial interpolation and the uniqueness of the interpolation polynomial.	Textbook and Lecture Handouts

	8	1	Stirling's interpolation formula	To determine the value of function from Stirling's interpolation.	Textbook and Lecture Handouts
		2	Bessel's interpolation formula	To determine the value of function from Stirling's interpolation.	Textbook and Lecture Handouts
		Mid Term Examination			
	9	1	Divided Differences, Newton's divided difference formula	Divided differences is an algorithm, historically used for computing tables of logarithms and trigonometric functions	Textbook and Lecture Handouts
		2	Numerical Differentiation Numerical Differentiation using Newton's Forward difference formula	An introduction to numerical differentiation and Numerical Differentiation using Newton's Forward difference formula.	Textbook and Lecture Handouts
	10	1	Numerical Differentiation using Newton's backward difference formula	Understanding to Numerical Differentiation using Newton's backward difference formula	Textbook and Lecture Handouts
		2	Numerical Differentiation using Stirling's formula	Understanding the Numerical Differentiation using Stirling's formula	Textbook and Lecture Handouts
	11	1	Numerical Differentiation using Bessel's formula	Understanding the Numerical Differentiation using Stirling's formula	Textbook and Lecture Handouts
		2	Lagrange's method	This formula is particularly useful for interpolating the values of f(x) for unequally spaced data.	Textbook and Lecture Handouts
	12	1	Numerical Differentiation Based on Lagrange's method	Understanding the Numerical Differentiation Based on Lagrange's method Students will be able to do the	Textbook and Lecture Handouts

		2	5 point method	Understanding the Numerical Differentiation Based on 5 point end point method.	Textbook and Lecture Handouts
	13	1	Numerical Integration Trapezoidal Rule with error analysis,	Numerical integration. Trapezoidal rule approximates the definite integral using trapezoidal approximations.	Textbook and Lecture Handouts
		2	Composite Trapezoidal Rule	Composite Trapezoidal rule approximates the definite integral using trapezoidal approximations Composite Simpson's rule approximates the definite integral by first approximating the original function using piecewise quadratic functions.	Textbook and Lecture Handouts
	14	1	Simpson’s 1/3 Rule with error analysis	Simpson's rule approximates the definite integral by first approximating the original function using piecewise quadratic functions	Textbook and Lecture Handouts
		2	Composite Simpson’s Rule	Simpson's rule approximates the definite integral by first approximating the original function using piecewise quadratic functions	Textbook and Lecture Handouts
	15	1	Simpson’s 3/8 Rule with error analysis	Understanding the difference between Simpson’s for even and odd intervals.	Textbook and Lecture Handouts
		2	Richardson Extrapolation	Students will understand the Richardson extrapolation.	Textbook and Lecture Handouts
	16	1	Introduction to Maple	An introduction to maple	Lecture Handouts
		2	Use of Maple programming	An overview of the use of maple.	Lecture Handouts

		Final Exam			

Semester Calendar for Assignments/Quizzes/Project

Assignments/ Projects and quizzes Plan	Week #	Assignment No.	Quiz No.	Project Title	Assignment/Project Quiz Date	Result Date of Assignment/Project/ Quiz
	1					
	2					
	3	Assignment-1				
	4		Quiz-1		Assignment-1	
	5					Assignment-1, Quiz-1
	6	Assignment-2				
	7		Quiz -2		Assignment-2	
	8					Assignment-2, Quiz -2
	9					
	10					
	11	Assignment-3				
	12		Quiz 3		Assignment-3	
	13	Assignment-4				Assignment-3, Quiz-3
	14		Quiz-4			
	15				Assignment-4	Assignment-4, Quiz-4
	16					

Instructor Name: **Zoobia Fatima**

Instructor Signature: _____

Date: